

An Introduction *to* Kubevirt, the Kubernetes Virtualisation Operator

Kubevirt is a Kubernetes add-on that enables the management of libvirt virtual machines. It is a tool that is used to schedule and manage virtual machines on a Kubernetes cluster.



With the ever growing adoption of containers in core deployment practices, the ease of porting applications becomes important. Kubernetes is becoming a front-runner in the containerised deployment framework with its feature set and awesome community support. In case the application cannot be containerised, we still should be able to use common frameworks to manage different workloads. An operator is a method of packaging, deploying and managing a Kubernetes application, which is anything that is both deployed on Kubernetes and managed using the Kubernetes APIs and kubectl tooling. You can think of an operator as the runtime that manages this type of application on Kubernetes.

The Operator Framework is an open source project that provides developer and runtime Kubernetes tools, enabling you to accelerate the development of an operator. This

framework includes the following:

The Operator SDK: This is to build operators based on their expertise, without requiring any knowledge of Kubernetes API complexities. It provides the tools to build, test and package operators. Initially, the SDK facilitates the marriage of an application's business logic, e.g., how to scale, upgrade, or back up with the Kubernetes API to execute those operations.

The Operator Lifecycle Manager: This oversees installation and updates as well as manages the life cycle of all the operators (and their associated services) running across a Kubernetes cluster. The Operator Lifecycle Manager is the backplane that facilitates the management of operators on a Kubernetes cluster. With it, administrators can control what operators are available in particular namespaces and who can interact with the running operators.